



Advanced Traffic (xx-xxx)
Class Project Part 6

Part 6: SUMO Simulation with the Same Configuration

Goal:

1. libsumo
2. Reproduce the Python-based traffic simulation using the SUMO traffic simulator.
3. Use python to generate everything.

Tasks

- Learn the basics of SUMO (network definition, route files, configuration files).
- Convert the previously defined network (nodes, edges, intersections) into SUMO input files.
- Generate vehicle demand based on the same logic used in the Python simulation.
- Run the simulation in SUMO and visualize traffic flow.
- Compare SUMO results with the custom Python simulation.

Output

1. SUMO configuration files (`.net.xml`, `.rou.xml`, `.sumocfg`)
2. Report file for comparison.
3. Presenting the code and running the file.

Time

Week 5-6

Grade Weight

25% (`.net.xml`: 5%, `.rou.xml`: 5%, run without error: 5%, correct outputs: 10%)